

Solution Design Document (SDD): **Smart Job Application Tracker**

Version History

Version Date	Description
v1.0	Initial Draft First version of Solution Design Document

Executive Summary

The **Smart Job Application Tracker** is a data-driven solution designed to help job seekers systematically track applications, analyze outcomes, and receive actionable guidance to improve hiring success. The solution combines application tracking, analytics, rule-based insights, and exploratory machine learning to transform raw job-search data into decision-support insights.

Purpose of the Solution Design Document

This document describes the **solution architecture, design decisions, workflows, and requirements** for implementing the Smart Job Application Tracker. It serves as a reference for development, validation, maintenance, and future enhancements.

Project Background

Job seekers typically manage applications using spreadsheets or notes, which provide limited visibility into trends, rejection reasons, and improvement areas. This project addresses that gap by introducing an intelligent tracking system that:

- Centralizes application data
 - Analyzes rejection patterns
 - Tracks interview progress
 - Provides learning and preparation recommendations
 - Explores ML-based success prediction
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Project Team

Role	Responsibility
Product Owner	Defines requirements and success criteria
Developer / Analyst	Design, development, analytics, ML implementation
Reviewer	Validation and feedback

Point of Contact

- **Project Manager (Client Side):** Job
 - **Technical Lead (Vendor Side):** Developer / Analyst (Project Owner)
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Business Requirements

- Track all job applications in a single system
 - Provide real-time metrics (total jobs, interviews, rejections)
 - Identify key rejection reasons and improvement areas
 - Support interview round tracking and preparation guidance
 - Enable search and querying of job data
 - Provide exploratory ML insights to estimate success probability
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Scope

In Scope

- Job application CRUD operations
- Analytics dashboards and visualizations
- Rejection reason analysis
- Interview round tracking
- ML-based success probability (exploratory)
- Search/chat-based querying

Out of Scope

- Guaranteed hiring outcomes
 - Enterprise-scale multi-user support
 - Automated job application submission
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Success Criteria

- User can track 100% of applications in the system
 - Dashboard metrics update correctly after every change
 - Rejection reasons are categorized accurately based on notes
 - Interview preparation guidance is generated for pending/failed rounds
 - ML metrics are visible and explained transparently
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Solution Overview

The solution uses a **hybrid approach**:

- **Rule-based analytics** for reliable insights with small datasets
- **Exploratory ML models** (Logistic Regression) for probability estimation
- **Search-based querying** for deterministic data retrieval

Note: While collaborative filtering is a common ML approach, this solution primarily uses **content-based analysis and behavioral patterns**, with ML as a learning layer rather than a recommendation engine.

Automation Flow

1. User adds or updates job application
2. Data is saved to CSV storage
3. Analytics refresh automatically
4. Rejection notes are parsed and categorized
5. Interview rounds are tracked and evaluated

6. ML model (if enabled) trains on available data
 7. User receives insights and improvement guidance
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Implementation Timeline

Phase	Duration
Requirements & Design	1 week
Core Development	2 weeks
Analytics & Visualization	1 week
ML Integration	1 week
Testing & Refinement	1 week

Technology Stack

- **Frontend:** Streamlit
 - **Backend:** Python
 - **Data Storage:** CSV (Pandas)
 - **Visualization:** Matplotlib
 - **Machine Learning:** Scikit-learn
 - **Optional AI:** LLM-based chatbot (API-driven)
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Current State Process Flow

1. Job seeker applies to jobs
 2. Tracks applications manually (spreadsheets/notes)
 3. No structured feedback or learning from rejections
 4. Repeated mistakes and low visibility
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Future State Process Flow

1. Job seeker records each application in the tracker
 2. System auto-generates metrics and trends
 3. Rejection reasons are analyzed
 4. Interview rounds are tracked with preparation guidance
 5. ML insights highlight improvement opportunities
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Functional Requirements

- Add, edit, delete job applications
 - Display dashboards and charts
 - Analyze rejection notes
 - Track interview rounds and statuses
 - Predict success probability (exploratory)
 - Provide learning and preparation recommendations
 - Support structured search queries
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Technical Requirements

- Python 3.9+
 - Streamlit runtime
 - Pandas for data manipulation
 - Scikit-learn for ML models
 - Environment-based configuration for API keys
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Assumptions and Considerations

- User enters accurate and detailed notes
- Dataset size is initially small

- ML predictions are exploratory
 - CSV storage is sufficient for single-user usage
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Risks and Dependencies

Risks

- Insufficient data for strong ML predictions
- Inconsistent or missing user notes
- Over-reliance on probability scores

Dependencies

- Python and library compatibility
 - Correct environment variable configuration
 - User discipline in maintaining data
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Results and Outcomes

1. Functional Results

- Successfully tracks all job applications in a centralized system.
- Real-time metrics display total jobs applied, interviews, and rejections accurately.
- CRUD operations (add/edit/delete) update analytics instantly without data loss.

2. Analytics Results

- Clear visibility into application trends (jobs per day, status distribution).
- Pie and bar charts effectively highlight pipeline health.
- Rejection analysis groups failures into actionable categories (resume, skills, experience, interview performance, visa/location).

3. Interview Preparation Outcomes

- Interview round tracking identifies cleared, pending, and failed rounds.
- The system provides targeted preparation guidance based on the next or failed round (e.g., Technical, Coding, Behavioral).

- Users can prioritize preparation areas instead of generic study.

4. Machine Learning Results (Exploratory)

- Baseline ML model implemented to estimate success probability (Interview/Offer).
- Evaluation metrics (Accuracy, ROC-AUC) are surfaced transparently.
- Results demonstrate current data limitations and highlight the need for richer features and more samples.
- ML is used as a decision-support tool rather than a deterministic predictor.

5. Search and Query Results

- Deterministic search enables fast filtering using structured queries (e.g., status:rejected, interviews>0).
- Users can retrieve relevant application subsets without manual scanning.

6. Business Value Delivered

- Converts raw job-search activity into measurable insights.
- Helps users identify recurring failure patterns and skill gaps.
- Reduces time spent tracking and increases focus on high-impact preparation.

7. Key Learnings

- Data quality and detailed notes significantly improve insight accuracy.
- Rule-based analytics provide reliable value with small datasets.
- ML performance improves as historical data volume and feature quality increase.

Conclusion

The solution delivers measurable improvements in visibility, learning, and preparation during the job search. While ML results are exploratory at the current stage, the overall system provides strong analytical and decision-support value, with a clear path for future enhancement.